Hyperbolic Polytope Volume as an LLM Benchmark Suite: Three Tasks of Increasing Freedom

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April 29, 2026

Abstract

We propose three complementary benchmark families for evaluating the mathematical reasoning capabilities of large language models on constructive geometry: (i) *inscribed reconstruction*, where the model is given the face lattice of a convex polytope and must produce vertex coordinates on the unit sphere realizing it; (ii) the *windowed-volume* task, where the model must construct an ideal polytope whose hyperbolic volume (or surface area or isoperimetric ratio) lies inside a narrow tolerance band around a specified target; and (iii) the *volume lower bound* task, where the model must report the sharpest lower bound it can on the maximum volume of an N-vertex polytope and back it with explicit vertex coordinates. The three tasks share a single grading principle—the answer must be a numerical certificate, evaluatable by a standalone checker—but they probe different reasoning regimes: pure combinatorial-to-metric inversion (i), control of a continuous objective (ii), and open-ended optimization with a falsifiable claim (iii). We report evaluations of three frontier models (Claude, GPT-5.4 Pro, Gemini 3.1 Pro DeepThink) on tasks (i) and (ii), revealing qualitatively different failure modes; task (iii) is released with full ground truth but the LLM evaluation is deferred. As a byproduct of the evaluation we record a combinatorial invariance phenomenon: the f-vector of the volume-maximizing polytope is the same whether volume is measured in the Euclidean or hyperbolic sense, across dimensions $d = 3, 4, 5, 6$.

1 Introduction

The rapid development of large language models has created a need for benchmarks that test genuine mathematical reasoning, not just pattern matching or memorization. An ideal benchmark should:

(i) have a *verifiable* answer that can be checked automatically;

(ii) require *construction*, not just classification or selection;

(iii) be *parameterizable*, so that memorization of specific instances does not help;

(iv) involve *non-trivial mathematics* that is unlikely to appear verbatim in training data.

The maximal-volume ideal polytope problem yields three distinct benchmark families that all satisfy these criteria. In each case, the model's response is a list of vertex coordinates on S^{d-1}, and the grader is a standalone Python checker that recomputes the relevant geometric measure and compares it to a target or record.

2 Three Benchmark Tasks

We summarize the three task families here; per-family results follow in §4.1, §4.2, and §4.3.

Task A: inscribed reconstruction. Given the f-vector, edge list, and facet list of a convex d-polytope, produce N points on $S^{d-1} \subset \mathbb{R}^d$ whose convex hull realizes that combinatorial type. This is a hard inverse problem: the model must find a point in a $(dN - \binom{d+1}{2})$-dimensional configuration space (after factoring out $\mathrm{O}(d)$) satisfying $O(N^2)$ combinatorial constraints—every listed facet must appear as a supporting simplex, and no extra facets may occur. Two challenge families are released: an *optimal family* drawn from our volume-maximizing inscribed polytopes (high automorphism order, often joins or Frobenius constructions), and a *random family* drawn from random points on S^{d-1} (generically trivial automorphism group, no exploitable symmetry).

Task B: windowed volume / surface area / ratio. Given integers N, d and a target T with tolerance ϵ, produce N points $v_1, \ldots, v_N \in S^{d-1}$ such that the ideal polytope $\mathrm{conv}(v_i) \subset \mathbb{H}^d$ has the chosen geometric measure (volume, surface area, or isoperimetric ratio V/A) within $[T - \epsilon, T + \epsilon]$. Targets are drawn from the 70th–95th percentile of the optimization landscape, so simple maximization is *insufficient*—the model must control the geometry to land in a specific window. The benchmark includes both hyperbolic and Euclidean variants, with $d = 4$ and $d = 6$, $N \in \{10, 20, 30\}$.

Task C: volume lower bound (anti-hallucination). Given (d, N) and a measure (Euclidean inscribed in the unit ball of $\mathbb{R}^d$, or hyperbolic ideal in $\mathbb{H}^d$), produce the sharpest lower bound the model can on the maximum volume, together with N explicit vertex coordinates that achieve it. The explicit-coordinate requirement is the anti-hallucination device: a model can easily quote a volume number from memory or produce a plausible-looking one, but it cannot easily fabricate a coordinate set that the standard convex-hull / Bloch–Wigner / Murakami routine evaluates to that number. The benchmark contains 94 challenges spanning Euclidean $d \in \{4, 5, 6\}$ and hyperbolic $d \in \{3, 4, 6\}$. Hyperbolic $d = 5$ is excluded by design: our stored $d = 5$ records at $N \in \{12, 14, 20\}$ are known local rather than global maxima, so grading against them would be unreliable in that dimension.

What the three tasks share. All three answers are graded by a Python checker (see §3) that returns exit code 0 for PASS and 1 for FAIL. All three are parameterizable: the challenge generators emit fresh (d, N) instances on demand. The mathematics underlying the verification—hyperbolic volume formulas, Murakami dilogarithms, Hopf reduction in $d = 6$—is sufficiently specialized that training-data contamination is unlikely.

What the three tasks probe differently. Task A is a pure combinatorial-to-metric inversion problem with a small number of degrees of freedom and many discrete constraints; it rewards algebraic recognition ("this is the join of two pentagons") as well as numerical constraint satisfaction. Task B asks the model to control a continuous objective, a strictly harder test than maximization—a model that can only maximize cannot pass. Task C is open-ended optimization with a falsifiable claim: any LLM that beats our record produces new mathematical knowledge.

3 The Verification Pipeline

3.1 Volume formulas

For $d = 4$, the volume of an ideal 4-simplex $[\infty, v_1, v_2, v_3, v_4]$ in the upper half-space model is given by the Haagerup–Munkholm formula:

$$\mathrm{Vol}_4 = \frac{\pi}{3}\left(4\pi - \sum_{i<j}\varphi_{ij} - \sum_{\ell}\arccos\frac{d_\ell}{R}\right),$$

where φ_{ij} are the Euclidean dihedral angles of the tetrahedron $T = \mathrm{conv}(v_1, \ldots, v_4)$, R is its circumradius, and d_ℓ are signed distances from the circumcenter to the faces.

For $d = 6$, ideal simplices have vanishing vertex links ($w_0 = 0$), and the Hopf reduction gives:

$$\mathrm{Vol}_6(\Delta_{\mathrm{ideal}}) = \frac{8\pi^3}{15}\left(\frac{17}{4} - w_4 + \frac{1}{2}w_2\right),$$

where $w_4 = \frac{1}{2\pi}\sum_{i<j}\theta_{ij}$ involves the 21 dihedral angles, and $w_2 = \frac{1}{2\pi^2}\sum_{|I|=4}\mathrm{Vol}_{S^3}(L_I)$ involves the 35 spherical tetrahedral link volumes, each computable by Murakami's dilogarithm formula.

3.2 The standalone checker

The checker (`standalone_checker.py`) is a self-contained Python script (no external dependencies beyond `numpy` and `scipy`) that verifies:

1. All N vertices lie on S^{d-1} (tolerance 10^{-6}).
2. All vertices are on the convex hull (no interior points).
3. The convex hull has the prescribed number of facets (if specified).
4. The computed geometric measure falls within $[T - \epsilon, T + \epsilon]$.

Exit code 0 indicates PASS; exit code 1 indicates FAIL. The checker supports seven measure types: `volume` ($\mathbb{H}^4$), `surface_area` ($\mathbb{H}^4$), `ratio` ($\mathbb{H}^4$), `volume_H6`, `euclidean_volume`, `euclidean_area`, and `euclidean_ratio`.

4 Model Evaluation

We evaluated three frontier models—Claude (with the toolkit), OpenAI's GPT-5.4 Pro (extended reasoning, "xhigh" effort), and Google's Gemini 3.1 Pro DeepThink—on Tasks A and B. Task C was designed after the bulk of this evaluation; its model results are deferred to future work and discussed in §4.3.

4.1 Inscribed reconstruction (Task A)

We tested GPT-5.4 Pro and Gemini 3.1 Pro DeepThink on instances of varying difficulty across the optimal and random families.

The two models employ qualitatively different strategies, as revealed by their chains of thought.

Table 1: LLM performance on inscribed polytope reconstruction. ✓ = correct combinatorial type, all points on S^3. × = wrong type or timeout.

N	Family	|Aut|	Facets	GPT-5.4	Gemini 3.1
10	optimal	200	25	—	✓
12	random	1	36	✓	× (crash)
20	optimal	16	72	✓	✓
30	optimal	1	124	✓	—

Gemini: algebraic recognition. For the $N = 10$ optimal polytope, Gemini identified the non-edge structure as two disjoint 5-cycles, recognized the polytope as the join $C_5 \oplus C_5$ of two regular pentagons in orthogonal 2-planes, and wrote down the explicit coordinates $v_j = (\cos 2\pi j/5,\ \sin 2\pi j/5,\ 0,\ 0)$ and $w_j = (0,\ 0,\ \cos 2\pi j/5,\ \sin 2\pi j/5)$. This is mathematically correct and produces an exact inscribed realization. However, when faced with the $N = 12$ random instance (trivial automorphism group, no recognizable structure), the same model crashed without producing output.

GPT: numerical constraint satisfaction. For the $N = 12$ random polytope, GPT's chain of thought reveals a direct numerical approach: "I've reformulated the search around determinant-sign tests on the tetrahedral cliques—enforcing same-side for the 36 listed facets and failure for the others." The model effectively runs a gradient-based optimizer over unit vectors on S^3, iteratively adjusting coordinates until the convex hull matches the target combinatorial type. This succeeds even for asymmetric polytopes with no algebraic shortcuts.

GPT on $N = 20$ and $N = 30$. The most striking results are GPT's successes on the larger optimal polytopes. At $N = 20$ (|Aut| = 16, 72 facets, 54 continuous DOF after symmetry reduction), GPT's chain of thought reveals a multi-pronged strategy: it first checked for known polytope types (24-cell subsets, 600-cell subsets), analyzed vertex figures, computed graph automorphisms, searched the Firsching [5] archive of inscribed polytope realizations, and ultimately fell back on numerical constraint satisfaction via determinant-sign tests on facet cliques. At $N = 30$ with trivial automorphism group (|Aut| = 1, 124 facets, 80 effective continuous DOF), no symmetry reduction is possible, and GPT's success demonstrates that its in-context numerical engine can navigate an 80-dimensional landscape with 124 nonlinear constraints with no algebraic shortcuts. Our own optimizer requires 200 random restarts of L-BFGS-B over the same configuration space to reliably find the $N = 20$ instance.

Gemini on $N = 20$. Gemini also solved the $N = 20$ optimal instance with a hybrid strategy: after failing to identify the polytope as a known type (24-cell subset, snub 24-cell, join), it computed the orbit structure under the automorphism group, identifying four orbits of sizes $[4, 4, 4, 8]$ and the antipodal pairs within each orbit. It then parameterized the configuration using orbit representatives, reducing the effective DOF before running L-BFGS-B with facet hyperplane constraints. This symmetry-reduced numerical approach is more sophisticated than a brute-force search over all $3N - 10$ parameters.

Discussion. For highly symmetric instances, Gemini's algebraic recognition (as on $N = 10$) provides an elegant shortcut. For asymmetric instances, GPT's direct numerical search (as on $N = 12$ random) succeeds where Gemini crashes. At intermediate difficulty ($N = 20$ optimal, |Aut| = 16),

both models converge via numerical methods, but with different parameterizations. The inscribed constraint is essential: without it, the problem reduces to combinatorial realization (finding *any* coordinates with the right combinatorics), which is substantially easier. Indeed, Gemini solved the non-inscribed $N = 10$ case using non-cospherical coordinates (norms 2 and $\sqrt{5}$) before we added the S^3 requirement. The hardest instances should be large N with both high volume and low symmetry—the $N = 16$ optimal polytope ($|\mathrm{Aut}| = 2$, 53 facets, volume per vertex 0.49) is a natural candidate.

4.2 Windowed volume / surface area / ratio (Task B)

4.2.1 Summary of results

Table 2: Model performance on selected benchmark tasks. Bold indicates the best answer. A dash indicates no numerical answer was produced.

Task	Best known	Claude	GPT-5.4 Pro	Gemini 3.1
$\mathbb{H}^4$ vol, $N = 30$	22.262	**22.262**	22.107	—
$\mathbb{H}^4$ vol, 20 simpl. facets	2.3139	**2.3139**	2.3139	1.61*
Eucl. vol, $N = 30$ on S^3	2.4909	**2.4909**	2.4839	—
Eucl. vol, 20 facets on S^3	0.7925	**0.7925**	**0.7925**	**0.7925**†
Eucl. area, 20 facets on S^3	5.9544	**5.9544**	**5.9544**	**5.9544**†
Eucl. area, 17 facets on S^3	5.036	5.002	**5.036**	—
$\mathbb{H}^6$ vol, $N = 16$	0.5406	**0.5406**	0.480‡	0.041*

*Gemini's answer is formally correct but useless (off by a factor of 13 for $\mathbb{H}^6$; stacked-simplex bound for 20 facets).
†Gemini provided the correct algebraic formula but no vertex coordinates, and falsely claimed numerical optimization confirmation.
‡GPT-5.4 Pro restricted to centrally symmetric polytopes ($\pm v_i$) and used QMC volume evaluation; removing the symmetry constraint and using the exact Hopf reduction formula recovers the full 0.5406.

4.2.2 Qualitative comparison

The three models exhibit qualitatively different strengths and failure modes.

GPT-5.4 Pro: persistent numerical constructor. Given a problem, GPT-5.4 Pro typically produces explicit coordinates, verifies them by multiple methods (e.g., star triangulations from all vertices), and states honest caveats ("I have not proved this is optimal"). It identified the free sum $C_5 \oplus I \oplus I$ as the Euclidean volume maximizer for 20 simplicial facets and derived the exact formula $V = \frac{5}{24}\sqrt{10 + 2\sqrt{5}}$. Its main weakness is occasional misinterpretation of the problem (e.g., reading $\mathbb{H}^6$ as the 6-cube in one instance, despite correctly interpreting $\mathbb{H}^4$ in a dozen previous conversations), which is easily corrected by re-prompting. On the 17-facet surface area problem, GPT-5.4 Pro found a configuration (area 5.036) that our 1000-restart optimizer could not beat.

Gemini 3.1 Pro DeepThink: strong algebraic reasoner, unreliable numerical claims. Gemini independently derived the exact formula $A = \frac{5}{3}\sqrt{15 - \sqrt{5}}$ for the 20-facet surface area, and proposed creative constructions (the 10-10 duoprism on the Clifford torus for 20 non-simplicial facets). However, it claimed that "random-initialization geometric optimization confirms" a numerical value of 12.4417 for the $N = 30$ surface area; when asked to produce the optimization code,

it spent over an hour and delivered code that only reaches 12.423. When confronted about the discrepancy, it entered a computation that ran for over 16 hours with no output. Its $\mathbb{H}^6$ answer ($4v_6 < M < 220v_6$) is a correct but useless interval spanning a factor of 55. On a separate occasion, Gemini produced a long, beautifully structured essay that misunderstood the optimization direction entirely (minimizing the supremum instead of maximizing the volume).

Claude (with toolkit): brute-force optimizer with certified answers. Using multi-start L-BFGS-B with 500+ restarts and a standalone verification pipeline, Claude's toolkit provides the strongest numerical results on most tasks but lacks the algebraic insight that GPT-5.4 Pro and Gemini occasionally provide. The hybrid approach—using Euclidean warm starts followed by hyperbolic polishing—was discovered during this evaluation and reduces optimization time by 38%. Several code improvements were adopted from the other models' suggestions: the Cartesian parameterization with norm projection (from Gemini 3.1's basin-hopping code) and the Powell optimizer for facet-constrained problems (from GPT-5.4 Pro's code).

4.2.3 The "windowed" benchmark format

A key design choice is the *windowed* format: the target value T is chosen from the 70th–95th percentile of the optimization landscape, not the global maximum. This prevents the model from simply maximizing the objective (which would give the same answer for every instance) and instead requires it to *control* the geometry to hit a specific window. This is substantially harder: a model that can only maximize cannot pass a windowed test with a target below the maximum.

We generated 66 windowed benchmark cases (33 hyperbolic, 33 Euclidean) covering volumes, surface areas, and isoperimetric ratios for $N = 10, 20, 30$ vertices, including facet-count-constrained variants.

4.2.4 Lessons for LLM training

The evaluation suggests that current training procedures produce characteristically different failure modes:

- **GPT-5.4 Pro** appears trained to minimize type II errors (failure to produce useful output), resulting in persistent numerical construction with honest uncertainty quantification.
- **Gemini** appears trained to minimize type I errors (false specific claims), resulting in a tendency to produce formulas without evaluating them, and to fabricate numerical confirmation when pressured for specific answers.
- The ability to produce *verifiable certificates* (code plus vertex coordinates) separates genuine computation from confabulation. Our checker enforces this distinction automatically.

4.3 Volume lower bound (Task C)

The volume-lower-bound benchmark was designed after the bulk of the windowed-volume evaluation above, principally as an anti-hallucination redesign: the windowed benchmark grades the submitted geometry against a target window and so a model that produces a plausible-looking number but no coordinates fails mechanically; Task C makes the requirement explicit by demanding the coordinates as part of the answer and grading the LLM's claim against what its own coordinates compute to.

Grading. The Task C grader is a single Python script that recomputes the volume of the submitted polytope under the appropriate measure (`ConvexHull` for Euclidean, the closed-form Bloch–Wigner / Haagerup–Munkholm / Murakami routines for hyperbolic), checks consistency between the LLM's claimed bound and the recomputed value within 10^{-3}, and compares the recomputed value to our internal record:

- **strong:** computed $\geq$ record (the model has matched or beaten our published bound; new mathematical claim);
- **pass:** computed $\geq 0.95\times$ record;
- **fail:** valid polytope but $< 0.95\times$ record;
- **invalid:** no coordinates, wrong dimension, or non-positive volume.

Coverage. The benchmark contains 94 challenges across five regimes:

Measure	Dimension	N values (count)
Euclidean inscribed	$d = 4$	5–24 (20)
Euclidean inscribed	$d = 5$	6–20 (15)
Euclidean inscribed	$d = 6$	7–24 (18)
Hyperbolic ideal	$d = 3$	4–300 (17, sparse at large N)
Hyperbolic ideal	$d = 4$	5–30 (14)
Hyperbolic ideal	$d = 6$	7–24 (10)

Hyperbolic $d = 5$ is excluded by design: our stored configurations at $N \in \{12, 14, 20\}$ are local rather than global maxima of the Euclidean (and hence hyperbolic) objective in $d = 5$, so grading against them would be unreliable for those instances and artificially easy at $N \in \{7, 8, 9, 10, 16\}$ where the records do match the global maxima.

Status. The benchmark files (`benchmark.json`, `ground_truth.json`, README) and grader (`check_volume_lb_response.py`) are released alongside this note. Frontier-model evaluation on Task C is deferred to future work; the design and ground-truth pipeline are reported here so the benchmark can be used by other groups.

5 Mathematical Byproducts

The LLM evaluation process, combined with systematic computational exploration, led to several mathematical discoveries.

5.1 Combinatorial invariance

The f-vector of the volume-maximizing ideal polytope is the same whether volume is measured in the Euclidean or hyperbolic sense. This was verified in $d = 3$ ($N \leq 50$), $d = 4$ ($N \leq 30$), and $d = 6$ ($N \leq 24$), with no exceptions. The invariance holds both within the regime where optimal polytopes are product polytopes and beyond it, suggesting a universal phenomenon.

5.2 Identification of optimal polytopes

For small N in even dimension $d = 2k$, the volume-optimal polytope is the join of k regular polygons in orthogonal 2-planes: $C_{p_1} \oplus \cdots \oplus C_{p_k}$. This is optimal up to $N \approx 2d + 2$, beyond which asymmetric configurations take over. Notable cases include the cross-polytope ($N = 2d$) and the simplex ($N = d + 1$). Volume-optimal polytopes are not cyclic polytopes and are in fact far from neighborly: at $N = 30$ in $d = 4$, the optimal polytope has only 123 facets compared to 756 for the cyclic polytope $C(30, 4)$.

5.3 The volume plateau

At $N = 9$ in $d = 4$ (equivalently, 20 simplicial facets), the hyperbolic volume maximum is achieved on a manifold of dimension ≥ 2, not an isolated point. The Euclidean maximizer (the free sum $C_5 \oplus I \oplus I$, with exact volume $\frac{5}{24}\sqrt{10 + 2\sqrt{5}}$) and the hyperbolic maximizer are metrically distinct configurations with identical hyperbolic volume to 13 significant digits. For prime facet counts (e.g., 17), the maximum is isolated.

5.4 Volume–volume correlation

Across the optimization landscape, the Euclidean and hyperbolic volumes are correlated at $r > 0.995$, with log-log elasticity approaching 2 at $N = 30$ in $d = 4$. This near-proportionality is surprising given that the Euclidean volume is polynomial in the vertex coordinates while the hyperbolic volume involves the transcendental Haagerup–Munkholm formula. For random configurations (as opposed to near-optimal ones), the correlation is lower ($r \approx 0.86$ in $d = 6$) but still substantial, and holds even within a fixed combinatorial type ($r \approx 0.88$ within a single facet-count bucket).

5.5 Distribution of random polytope volumes

For N points sampled uniformly on S^{d-1}, both the Euclidean hull volume and the ideal $\mathbb{H}^d$ polytope volume are random variables whose distribution depends on N and d. In $d = 6$, we find a transition in the distribution shape as N grows:

N	Skewness	Normal p	Beta p	Best fit
10	$+0.78$	< 0.001	0.985	Beta$(3.9, 43.7)$
12	$+0.65$	0.005	0.766	Beta$(6.9, 120.7)$
16	$+0.23$	0.088	< 0.001	Borderline
20	$+0.02$	**0.832**	< 0.001	Normal
30	-0.09	0.281	< 0.001	Normal
50	-0.19	0.034	< 0.001	Normal (slight left skew)

For small N (≤ 12), the volume distribution is well-described by a Beta distribution with parameters scaling linearly in N, consistent with earlier observations in $d = 3$ and $d = 4$. For large N (≥ 20), the distribution converges to Normal—as expected from a CLT-type argument, since the polytope volume becomes a sum of many weakly-dependent simplex volumes in the Delaunay triangulation. The crossover occurs at $N \approx 16$–20 in $d = 6$ (where the skewness crosses zero), and is accompanied by a subtle transition from right skew to slight left skew.

The within-bucket distributions (conditioning on a fixed facet count) show essentially the same shape and correlation structure, confirming that the distributional properties are genuinely geometric rather than combinatorial artifacts.

6 Conclusions

The three benchmark families above expose qualitative differences between frontier LLMs that traditional math benchmarks miss. The ability to produce *verifiable certificates*—explicit coordinates that pass an independent numerical checker—is a fundamentally different capability from producing plausible-sounding mathematical text. The three tasks probe different capabilities: inscribed reconstruction (Task A) is a constraint-satisfaction problem rewarding both algebraic recognition and numerical search; the windowed-volume task (Task B) requires controlling a continuous objective rather than just maximizing it; and the volume-lower-bound task (Task C) is anti-hallucination by design—a numerical claim that does not match the model's own coordinates is automatically rejected.

All three benchmarks are parameterizable (any N, d, and either target window or measure), resistant to memorization (windowed targets from a continuous landscape; record values withheld in Task C), and fully automatically gradable (standalone checkers with exit codes). The mathematical content—hyperbolic volume, Murakami dilogarithms, Hopf reduction, Bloch–Wigner—is sufficiently specialized that training-data contamination is unlikely.

As a byproduct, the evaluation process itself generated new mathematical results (combinatorial invariance, identification of optimal polytopes, volume plateau), demonstrating that the "mixture of experts" approach —combining human mathematical intuition with LLM algebraic insight and automated numerical optimization—can be more productive than any single participant acting alone.

References

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